F43 adaptive S3: setup and target request accounting

Quadratic-twist study, PR2381

25 October 2026; phase-demand-20261025

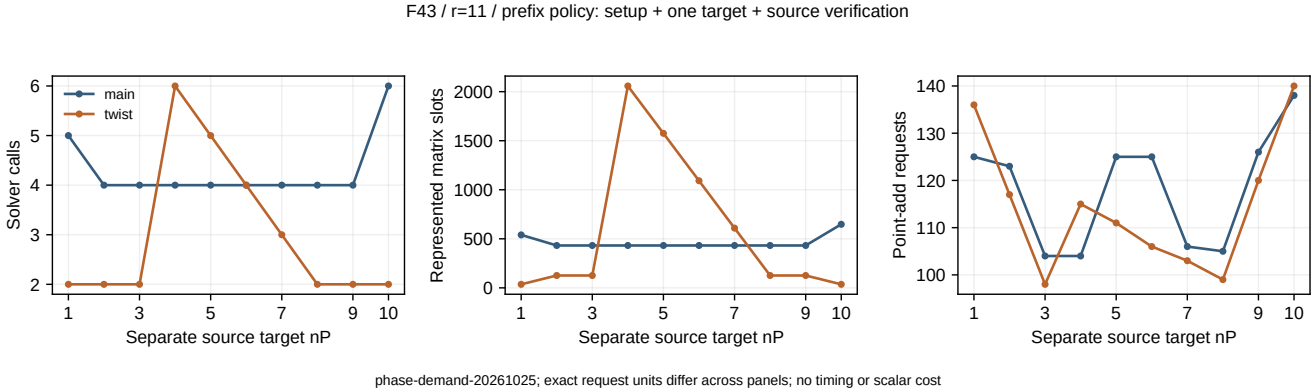
Executed experiment. Native C++ and independent Rust replay 40 base setups and 400 separate nonidentity source-point recoveries: ten frozen policies, four roles, ten source points per role/policy. All 400 scalars verify on the main curve. Every phase counter, empty query, sign lift and failed shift is retained. A distinct batch-RREF/witness check verifies the 100 unique solver certificates used. This is an accounting result, with no timing or scalar-cost claim.

The first policy shows the tradeoff

Use $E : y^2 = x^3 + x$, $T : Y^2 = X^3 + 4X$ over F_{43}, order-11 generators $(31,18)$ and $(24,26)$, and bases $(4,5),(13,19)$ and $(15,9),(17,6)$. Existing models are `icv1-fp6-t0-74779950` and `icv1-fp6-t0-c6a8a47c`. For the separate source $Q = P$:

Request unit	Main	Twist
Setup solver calls	3	1
Online solver calls, failures included	2	1
Cold solver calls	5	2
Represented matrix column slots	540	36
Point-add requests	125	136
Membership-polynomial evaluations	645	258
Coordinate-map multiplications	0	2

The previously verified degree/rank difference is retained. The new ledger shows that fewer solver requests and matrix slots coexist with more actual point-lifting requests. The graph retains all ten source targets.



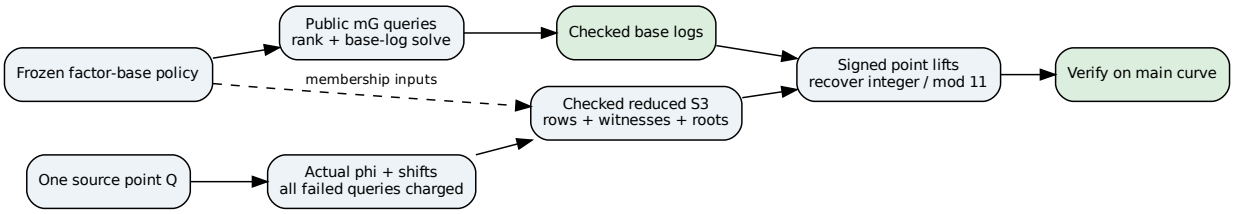
Exact native requests in three distinct units; not times or equal-cost operations. Each point includes its policy setup, one source target and source verification.

All policies and transported controls

Across 100 unmatched policy-target comparisons, the twist's cold solver calls are lower/equal/higher in 35/36/29 cases, represented slots in 44/15/41 and point-add requests in 49/1/50. Both presentations have summed hypothetical cold solver requests 393 and slots 75,804. These sum separate records, with no multi-target speed average. The main-to-twist matched base and inverse-base control reproduce every request component except two coordinate-map multiplications per nonidentity input. Repeated policies are not independent curve trials or probabilities.

Exact request model

The 29 counters include root membership and constraint evaluations, every matrix generation and lower-degree membership rebuild, polynomial and witness term copies/updates, pivot normalizations, numeric evaluation and subgroup linear-system row operations, scalar iterations, group-law requests, sign lifts, membership/remainder construction and recovery requests. Dense column slots describe represented rectangular demand; sparse code is not claimed to allocate it. Some counters are nested. They cannot be added into a calibrated time or GAE. Hashing, fixed field-table setup, allocation, sorting, traversal, internal exponentiation and independent audit remain unpriced.



Lean bridge to source recovery

For an injective additive map ϕ, explicit base equations $B_0 = l_0\phi(P)$, $B_1 = l_1\phi(P)$, and a signed point relation

$$iB_0 + jB_1 = \phi(Q) + s\phi(P),$$

Lean proves $Q = (il_0 + jl_1 - s)P$. Reducing the integer modulo r explicitly uses $rP = O$. Five new theorems in `lean-phase/PhaseRecovery.lean` include the actual F43 annihilator $11P = O$ and instantiate the argument on the frozen injective additive map $\phi(x,y) = (41x, 11y)$. This exceptional pair is base-field-isomorphic; no general rational-point twist map is assumed. The native audit checks the base equations, signed relation and source scalar in every recorded call. The full native algorithm is not claimed formalized.

The unmodified scoped gate passes 176 public research theorems and 231 audited declarations, zero failing/extra trusted axioms. All fourteen prior polynomial certificates and sources remain unchanged. Six rehashed controls reject false setup totals, matrix demand, omitted failed query, base log, answer and source binding. Exact replay and mandatory library/backend/metadata checks are linked in `validation/phase-demand-20261025`.

Next discriminating implementation

Each solver query here performs 129 membership evaluations before four reduced-S3 checks. A proposed next implementation supplies the already verified two factor-base roots directly and proves exact constrained-root equivalence before re-running the same accounting workload. This optimization has not executed in this report. Low-level pricing and appropriate references remain required for a speed claim. Shared Conductor, current-head CI/review, integration and broader geometric/completion gates remain open.